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Monolithic 6-Axis Force/Torque Sensing by a Single Ultrathin Piezoceramic Shell

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Abstract

Miniature multi-axis force/torque sensors are essential for advanced robotics and wearable devices, yet their development is hindered by complex multi-beam structures and laborious calibration. We overcome this by introducing a monolithic 6-axis force/torque sensor constructed from a single, ultrathin-walled piezoceramic shell. This architecture eliminates the need for multi-component assembly and algorithm-driven decoupling, relying instead on the intrinsic mechanical deformation modes of the engineered shell for direct and reliable dynamic force/torque variation measurement. Leveraging this architecture, our sensor enables straightforward calibration and achieves a decoupling accuracy of 99.37% across 120,000 randomized dynamic tests. We fabricated the core sensing element as a 50-μm-thick annular piezoceramic shell to maximize sensitivity and limit of detection, achieving detection limits better than 3 mN in normal force, 4 mN in tangential force, and 0.3 mN·m in torque. We demonstrate a compact, 3.85-gram sensor (2.53 cm^3) that integrates seamlessly into robotic grippers for event-driven delicate assembly and into exoskeletons for high-fidelity dynamic monitoring of in-home rehabilitation. This work establishes an architecture for inherently decoupled force/torque sensing, with broad potential in advanced robotics, prosthetics, and personalized medicine.

Keywords: Monolithic; 6-Axis Force/Torque Sensor; Soft Machined Piezoceramic Shell; Robotic Collaboration; Wearable Rehabilitation.

Introduction

The ability to decode multidimensional dynamic loads is crucial for the next generation of intelligent machines, enabling robots to perform dexterous manipulation and wearable devices to deliver personalized interventions[1-3]. While 6-axis force/torque (F/T) sensors provide the full spectrum of load characterization[4,5] beyond the capacity of 3-axis force sensors[6-13], their broader adoption remains limited by a persistent architectural constraint. Conventional designs, though commonly used in robotic joints and industrial settings[14,15], are typically constructed from multiple structural beams instrumented with an array of sensing elements. This configuration inherently introduces excessive structural rigidity, complicates signal routing, and requires extensive calibration. Consequently, a fundamental trade-off between miniaturization and performance has long persisted[16-18], restricting their use in applications such as robotic fingertips or wearable medical systems[19,20].

Recent efforts to miniaturize 6-axis F/T sensors have generally followed two divergent paths[21-25], both of which struggle with this trade-off. One approach retains the multi-beam sensing structure[26-28], which becomes increasingly difficult to fabricate and isolate the signals at small scales. The alternative employs soft or flexible sensing elements, some in dense arrays, which simplify fabrication but transfer the decoupling burden entirely to computational algorithms such as convolutional neural networks[29-31]. These data-driven methods lack interpretability, require large training datasets and complex algorithms, and often generalize poorly. Thus, a fundamental redesign of the sensing architecture, moving beyond multi-element assemblies and black-box data-driven decoupling, is essential to realizing truly miniaturized, high-performance F/T sensing.

Here, we present a monolithic 6-axis F/T sensor architecture based on a single, ultrathin piezoceramic shell. This design departs from the multi-beam architecture, integrating all sensing functions into a unified structural element. Force and torque separation is achieved through the shell's distinct mechanical deformation modes, which generate spatially differentiated charge distributions. An optimized electrode pattern effectively extracts orthogonal components through spatial and polarity-based signal differentiation, thereby significantly reducing the computational complexity of decoupling. This architecture-based approach significantly minimizes intrinsic crosstalk and enables straightforward calibration. We fabricate the core sensing element as a 50-μm-thick annular PZT shell, which provides exceptional sensitivity, yielding LODs better than 3 mN in normal force, 4 mN in tangential force, and 0.3 mN·m in torque. It achieves a decoupling accuracy of 99.37% across 120,000 randomized dynamic tests. We demonstrate its practical utility through seamless integration into a robotic gripper[32,33] for delicate assembly and a lightweight exoskeleton[34,35] for high-fidelity rehabilitation monitoring (Fig. 1a). This work provides a method for inherently decoupled multi-axis sensing, with applications in advanced robotics, prosthetics, and personalized medicine.

Results

Design and optimization of the F/T sensor

The decoupled 6-axis F/T sensing is achieved by capturing and classifying the strain behaviors of different regions on a piezoceramic shell. For this purpose, we propose a F/T sensing structure comprising an annular piezoceramic shell, a soft inner layer, and soft lid assembled together (Fig. 1b). When external loads act on the lid, they are transmitted through the inner layer to the PZT shell. Fig. 1b also illustrates the stress distribution in the PZT shell when the soft lid is subjected to positive 3-axis forces and 3-axis torques. Subsequently, different regions of the PZT shell will generate charges of opposite polarity under tensile or compressive stress due to the direct piezoelectric effect[36]. The results clearly demonstrate that the stress distributions in the PZT shell differ remarkably under various loads, which provides the basis for the proposed F/T sensing structure to realize directly decoupled 6-axis force/torque sensing. On the basis of this F/T sensing structure, we replaced the sensing element with a 50-μm-thick annular PZT shell and incorporated spike-shaped microstructures on the inner layer for enhanced sensitivity and detection limits (Fig. 1c). Additionally, we designed patterned outer electrodes for the PZT shell to visualize its stress distribution (by outputting charge signals), as well as for subsequent load decoupling. Thus, a monolithic 6-axis F/T sensor design is proposed. As shown in Fig. 1d, this design features compact size and light mass, as it relies solely on a single sensing element and utilizes the soft silicone encapsulation to replace traditional bulky metal housings. Meanwhile, the 50-μm-thick sensing element offers low limits of detection for normal force, tangential force, and torque. The detection capability stems from the enhanced charge response of the thinned annular PZT element (Fig. 1e) to slight loads, and this phenomenon is attributed to the mutation of elastic stiffness[37,38] in annular PZT devices below 100 μm in thickness (See more details in Supplementary Note 1).

Fabrication and characterization of the ultra-thin sensing element

A customized soft machining technology exploiting multi-segment milling strategy (refer to the section of 'Soft machining process for annular PZT sheets' in the Methods) has been developed to fabricate annular PZT shells of various diameters, heights and thicknesses with using sintered annular PZT shells (Fig. 2b). As shown in Fig. 2a, the soft cutter and the thick PZT shell counter-rotate for mechanical thinning (indicated by black arrows), and a gel milling fluid doped with silicon carbide is used to accelerate material removal. After processing, the outer electrodes of the machined PZT shell are re-patterned via a conductive silver paste printing process (Supplementary Fig. S5a). The entire machining process involves three stages: rough machining, finish machining, and polishing. Taking the machining process of a 20-mm-diameter annular PZT shell with a target thickness of 55 μm as an example (Fig. 2c), the entire process is visualized with intuitive milling force feedback, and the final thickness of the machined annular PZT shell is ~57 μm (Supplementary Fig. S3). Fig. 2d illustrates a comparison between the machined 57-μm-thick annular PZT shell and the 500-μm-thick annular PZT shell during processing. The reduced surface roughness of the polished PZT shell (Fig. 2e and Supplementary Fig. S5c) directly contributes to a smoother and more uniform printed silver electrode.

To evaluate the benefit of thickness variation on the response performance of annular PZT shells, several experiments focusing on contactless acoustic pressure detection were conducted. Fig. 2f illustrates the experimental setup, where a sound level meter serves as the reference for calibration. Fig. 2g displays the measured sensitivity of annular PZT shells across different thicknesses and excitation frequencies. The results confirm that the resonant frequency decreases with reduced thickness, while the peak sensitivity at the resonant frequency is markedly enhanced. Subsequently, by combining these results with the response sensitivity curves (Fig. 2g), the minimum detectable pressure (MDP) was determined. As shown in Fig. 2h, the measured MDP of a 50-μm-thick annular PZT shell reaches 9.43 μPa/√Hz. These results collectively demonstrate that the 50-μm-thick annular PZT shell exhibits high acoustic pressure sensitivity, highlighting its strong potential for applications in high-sensitivity F/T sensors.

Decouple mechanism and characterization of the annular F/T sensor

The designed sensor (Fig. 3a for exploded view) primarily consists of a 50-μm-thick annular PZT shell with specially patterned outer electrodes, a soft encapsulation layer, a soft inner layer with lid and spiky microstructures, and rigid support parts for protection as well as fixation (refer to the section of 'Fabrication and assembly procedures' in the Methods). As shown in Fig. 3b, the soft inner layer, PCB and base are tightly assembled together through a sleeve fit. Specifically, a spike shaped microstructure is incorporated into the inner layer to enhance the responsiveness to minute loads. Furthermore, a combination of high modulus silicones (Shore hardness 30A and 60A) is selected to ensure structural stability and guarantee the dynamic response to high-frequency loads. Fig. 3c shows the fabricated F/T sensors (with its key performance indicators summarized in Supplementary Table 2). Its sensing mechanism relies on the direct piezoelectric effect. As shown in Fig. 3d, once the inner layer is bonded to the annular PZT shell via the spikes, the PZT shell generates positive charges under compressive stress and negative charges under tensile stress transmitted from the inner layer. The outer electrodes of the PZT shell were specifically designed to capture and differentiate strain responses under various loading conditions. As shown in Fig. 3e and Fig. 3f, six distinct electrodes S_1 to S_6 are deployed with a common ground. Among them, S_1 and S_2 are aligned along the positive X-axis, S_3 and S_4 with the positive Y-axis, and S_5 and S_6 with the negative X-axis and Y-axis, respectively.

Fig. 3e illustrates the simulated charge response of electrodes S_1 to S_4 under various loading conditions, where blue and orange shades denote positive and negative charges, respectively.

Consistent with the structural mechanism shown in Fig. 1b, the distinct electrode outputs arise from the non-uniform stress distribution across the PZT shells under multi-axis loads. Each load component induces a distinct combination of signal patterns (complete FEA results for all 12 loading types are provided in Supplementary Fig. S8). For instance, under a $+F_x$ load, the positive X-axis displacement of the lid compresses the regions near S_1 and S_2 while pulling the remaining electrodes, resulting in opposite charge polarities. This electrode configuration offers significant advantages for decoupling by effectively capturing the geometric deformation response of the PZT shell. The annular geometry ensures that each load component induces a distinct structural response, resulting in high geometric orthogonality among signal vectors and an optimized condition number for the global decoupling matrix (refer to the section of 'Decoupling and optimization process' in the Methods). Furthermore, the bipolar nature of the charge signals increases information density per channel. This supports a deterministic mapping using linear matrices and polynomial compensation, achieving precise load resolution while circumventing the need for complex black-box algorithms. The integrated working flow is summarized in Fig. 3f. External mechanical stimuli induce structural deformations that generate independent charge signals across the six electrodes. These signals are processed through a metrology chain (refer to the 'Design of the piezoelectric sensor metrology chain' in the Methods) to yield measurable voltage outputs. Finally, the system utilizes the deterministic mapping of a global matrix and polynomial compensation to resolve the six-axis force and torque components. Building upon this operational framework, the baseline performance of the sensor was characterized under no-load conditions to determine the resolution limit. Fig. 3g displays the noise floor for each axis, where RMSE values for force and torque loads remain below 1.3 mN and 0.13 mN·m, respectively. These results confirm the high-fidelity acquisition capability for subtle mechanical inputs.

Subsequently, an experimental platform integrating a commercial 6-axis F/T sensor (Fig. 3c) was constructed to conduct the sensor calibration and performance characterization (more details can be found in Supplementary Note 4). Based on this setup, the systematic performance of the proposed sensor was comprehensively evaluated (refer to the section of 'Performance characterization of the designed F/T sensor' in the Methods). Fig. 4a illustrates the average response sensitivity of the sensor electrodes under dynamic loading ranging from 1 to 40 Hz (the complete sensitivity matrices across all 40 frequencies are detailed in Supplementary Figs. S12 and S13). Attributed to the rigid-soft hybrid architecture and the utilization of high-modulus silicone, the measured sensitivity matrices for the 6-axis loads remain highly stable across various frequencies. Notably, the maximum coefficient of variation for the matrix elements within the tested bandwidth is merely 1.78% (Supplementary Fig. S14). To further assess this dynamic stability, the variation range of the equivalent sensitivity for each load axis is presented in Fig. 4b, indicating a highly stable dynamic response within the testing bandwidth. Fig. 4c displays the average cross-axis crosstalk matrix under the same dynamic tests, expressed as a percentage of the full-scale output (% F.S.) of the corresponding axis. Compared to the primary sensitivities, the crosstalk coefficients are less susceptible to the loading frequency. Across the wideband evaluation, the maximum analytical crosstalk was recorded during the 31 Hz test; specifically, a full-scale load in the M_x direction induces a 2.38% F.S. error on the F_y axis (Supplementary Fig. S16).

Following calibration, the sensor demonstrated exceptional decoupling capabilities under dynamic conditions. Across a comprehensive evaluation comprising 120,000 random loading trials within the tested bandwidth (the detailed loading protocol is provided in Supplementary Table 7), the sensor achieved decoupling accuracies of 99.377% and 99.373% for force and torque loads, respectively (Fig. 4d). Moreover, for the correctly identified load types, the maximum magnitude error between the sensor outputs and the reference values remained strictly below 2.5% F.S. (Fig. 4e). Notably, it exhibits an exceptionally high accuracy particularly within the 1–15 Hz frequency range (complete measurement data are detailed in Supplementary Figs. S25 and S26). The limit of detection (LOD) for dynamic loads was also systematically evaluated (Fig. 4f). Within the tested frequency band, the proposed sensor exhibits an average LOD below 3.5 mN for force loads and 0.27 mN·m for torque loads. The impacts of operating temperature on the sensor were also characterized (refer to 'Effect of temperature variations on the sensor' in the Methods). Finally, the long-term reliability and fabrication reproducibility were strictly validated. As depicted in Fig. 4g, the sensor exhibits a high long-term stability, with negligible degradation in equivalent sensitivity observed over an 8-week continuous testing period. Moreover, the fabrication reproducibility is confirmed in Fig. 4h, where eight fabricated sensor samples demonstrate highly uniform sensitivities under 1 Hz dynamic loading (additional performance metrics are detailed in Supplementary Figs. S38 and S39).

F/T sensors for dexterous robotic gripper-arm collaboration

Featuring a compact architecture, the proposed sensor can be seamlessly integrated into a robotic gripper to provide 6-axis F/T tactile feedback (Fig. 5a). This integration empowers the gripper with comprehensive state information, particularly the relative motion tendencies of manipulated objects. The force-sensing capability was first validated through an adaptive clamping task handling a slippery beaker (Fig. 5b). As the beaker is gradually filled with water, gravity-induced slip occurs at the contact interface. By establishing a predefined threshold for the dynamic tangential force variation, the gripper can detect this incipient slip and automatically perform a corrective secondary clamp (Supplementary Video 1). The real-time raw 3-axis force variations monitored by the integrated sensor offer a highly intuitive and quantitative characterization of this entire self-adjusting process (see Supplementary Fig. S41 for the comparison with filtered signals).

To further elucidate the importance of implementing 6-axis F/T sensing on robotic gripper fingers, we demonstrate two robotic gripper-arm collaboration tasks, primarily based on fingertip torque feedback (further details on the closed-loop and collaboration methods are provided in Supplementary Note 7). The first task is the center of gravity (COG) estimation for a transparent glass tube. Recognizing the COG of transparent objects is challenging for machine vision methods due to distortions caused by optical reflections, thereby necessitating the assistance of multidimensional F/T sensing. As illustrated in Fig. 5c, the entire task can be divided into three stages: 1) From 0 s to 15 s, the gripper initially grasps the object at point A with a normal force of 4 N (approximating the critical force for grasping, calculated from the 200-gram object weight and a friction coefficient of 0.5), and then lifts it at a constant velocity. Since the grasping point is offset from the actual COG, the object slides, and the transient peak of the torque variation is recorded as M_{z1}. 2) From 16 s to 35 s, guided by the camera, the gripper moves to the opposite end and grasps the object at point B with the same force. Upon lifting at the same speed, the object slips again, and the corresponding peak torque variation is recorded as M_{z2}. 3) From 36 s to 43 s, based on the calculated COG, the gripper repositions to the exact center and successfully lifts the object with the same normal force without any observable slip (Supplementary Video 2, and see Supplementary Fig. S42 for the comparison with filtered signals). The COG calculation is performed via torque inversion, leveraging the relationship between the peak torque variation and the distance from the grasping point to the COG, which can be expressed as follows:

$$\mathrm{OA/OB}=M_{z1}/M_{z2} \tag{1}$$

The second task involves a simulated battery pack assembly (Supplementary Video 3), representing a typical high-precision industrial operation. During this process, the robotic system relies on multi-axis sensing feedback to continuously adjust its pose for accurate insertion. As illustrated in Fig. 5d, the task is divided into two main stages: 1) From 0 to 8 s, the gripper transports and aligns the battery pack. By sensing transient external impacts and orientational deviations during alignment, the system continuously corrects its pose to prevent assembly misalignment. 2) From 9 to 12 s, the gripper executes the insertion phase. Based on the real-time 6-axis F/T variations, the system fine-tunes the insertion trajectory to avoid excessive transient impact forces between the pack and the slot, thereby ensuring safe assembly. The corresponding dynamic F/T variations monitored by a single sensor are also plotted in Fig. 5d (see Supplementary Fig. S43 for the comparison with filtered signals). Overall, these results demonstrate that the proposed F/T sensor endows robotic systems with comprehensive 6-axis tactile perception. This multidimensional tactile feedback enables the precise execution of complex manipulation tasks, highlighting its potential to supersede conventional 3-axis tactile solutions.

F/T sensors enabling intelligent exoskeletons

To further demonstrate its applicability in wearable devices, the F/T sensors were integrated into an exoskeleton. As shown in Fig. 6a, the inner layer of the sensor is mechanically coupled to the spindle of an exoskeleton, directly transmitting the wearer's dynamic joint torque variations to the sensing element[39-41]. Fig. 6b illustrates the integration schematic of an intelligent exoskeleton equipped with two F/T sensors and an untethered metrology chain for data processing and transmission. Specifically, sensor 1 is mounted on the inner interface to monitor the transient interfacial friction (ΔF_x, ΔF_y) and compressive force variations (ΔF_z) between the wearer's leg and the exoskeleton during locomotion, while sensor 2 measures the equivalent joint torque variation (ΔM_z). This multidimensional kinetic monitoring effectively addresses the current lack of comprehensive dynamic kinetic feedback in conventional exoskeletons. The real-time 3-axis force variations from sensor 1 quantitatively reflect wearing comfort by capturing excessive collisions and micro-slips, guiding users to optimize their donning process (e.g., adjusting the tension of multidirectional cross-straps). Such sensor-guided adjustments allow users to achieve a self-adjusted, ergonomic fit using affordable standard exoskeletons (approx. $10), circumventing the prohibitively high costs associated with individualized morphological customization (up to $260). Finally, by correlating the multidimensional dynamic force data with user-reported experiences, optimal dynamic interaction ranges for an ergonomically sound fit were statistically derived (Supplementary Fig. S44e; detailed discussion in Supplementary Note 8).

Additionally, the equivalent joint torque variations measured by sensor 2 facilitate the rehabilitation assessment of patients with joint injuries. Since the dynamic knee joint torque variation reflects the transient resultant force of muscle contractions and closely correlates with joint angular acceleration, this relationship was quantitatively verified by simultaneously monitoring quadriceps activation via electromyography (as shown in Supplementary Fig. S47c). As depicted in Fig. 6c, the peak joint torque variations and peak EMG signals exhibit a strong positive correlation with joint angular acceleration during bent-leg tests (further details in Supplementary Fig. S47e). To establish evaluation baselines, peak joint torque variations were systematically collected from healthy participants with diverse body morphologies during activities of daily living, including stair climbing, uphill walking, elliptical exercise, and load carriage (Supplementary Fig. S47g). Building upon these baselines, the intelligent exoskeleton was deployed to monitor a wearer during a stair-descent task (Supplementary Video 4). As illustrated in Fig. 6d, the continuous signal trajectories intuitively map the varying motion states across three distinct stages: a slow descent (0–3 s), a fast descent (4–6 s) with correspondingly amplified signal frequencies, and a terminal jump followed by a landing (7–9 s). The measured contact force variations directly reflect the step frequency, whereas the measured equivalent joint torque variations indicate muscle exertion across different speeds (see Supplementary Fig. S46 for the comparison with filtered signals). Quantitative extraction of these parameters yields two insights. First, the peak right-knee torque variation reaches approximately 23 mN·m during rapid leg lifting (Fig. 6e), reflecting robust muscle activity and dynamic joint performance consistent with other healthy participants (Supplementary Fig. S47h). Second, the peak compressive and frictional force variations during leg lifting and landing reveal excessive leg-exoskeleton dynamic impacts (Fig. 6f). Notably, the transient compressive impact exceeds 5 N during the fast descent stage, surpassing the ergonomic comfort thresholds. This explicitly indicates that subsequent strap adjustments are necessary to prevent excessive dynamic mechanical pressure on the skin during prolonged wear.

Finally, the application of the integrated exoskeleton in rehabilitation assessment was demonstrated for users during their post-injury recovery. As illustrated in Fig. 6g, a typical joint injury not only alters the dynamic joint torque generation but also disrupts the transmission of leg-movement signals, thereby prolonging the time delay between muscle activation and actual joint movement. On this basis, an accessible in-home rehabilitation monitoring protocol was established, requiring only the intelligent exoskeleton, a surface EMG sensor, and basic exercise equipment, such as a stationary bike (details are provided in Supplementary Note 8 and Supplementary Fig. S48). Fig. 6h illustrates the measured peak joint torque variations and the EMG-to-torque time delay as a function of rehabilitation duration. Ultimately, the proposed exoskeleton-based assessment framework demonstrates the integration of daily physical training with precise clinical evaluation, offering a cost-effective approach that facilitates active recovery management for users.

Discussion

In summary, we have developed a monolithic 6-axis force/torque sensor architecture that achieves load decoupling through a single, engineered piezoceramic shell. This design eliminates the need for multi-element assemblies and complex algorithm fusion, offering a fundamentally simplified alternative to conventional F/T sensors. The core of this sensor, a 50-μm-thick annular PZT shell with optimized electrode patterning, enables high-sensitivity detection and intrinsic differentiation of loads, resulting in a low limit of detection and a decoupling accuracy exceeding 99.37%.

We demonstrated the practical versatility of this sensor by seamlessly integrating it into two distinct systems: as a fingertip tactile sensor for robotic grippers, and as a kinetic monitor for wearable exoskeletons. These integrated systems successfully performed advanced tasks, including the precise assembly of delicate components and the quantitative assessment of joint movement during in-home rehabilitation.

Looking forward, the architecture established here is not merely a sensor design, but a scalable platform for inherently decoupled multidirectional sensing. Finite element analysis suggests that the spatial resolution and off-axis load decoupling performance can be further enhanced through refined electrode segmentation and microstructure patterning. This principle enables the development of the next generation of high-performance, miniaturized force/torque sensors, with broad potential to advance fields such as collaborative robotics, intelligent prosthetics, and personalized healthcare.

Methods

Fabrication and assembly procedures.

The encapsulation and inner layer of the F/T sensor were fabricated using silicone molding processes. Supplementary Fig. S6a shows the fabrication of the soft layer with microstructures: first, the 3D-printed molds are assembled, and uncured PDMS is poured for the initial molding step. After heating and curing, the PDMS mold is prepared. Subsequently, it is assembled with the heat-resistant resin part, and 30A silicone is poured for the second molding step. Following high temperature heating and curing, the soft inner layer with microstructures is completed. Notably, the tips of the microstructures are curved to ensure better bonding to the inner surface of the annular PZT shell. Supplementary Fig. S6b illustrates the integrated fabrication of the soft lid and inner layer. First, a customized mold featuring a regular array of internal conical recesses was designed. These recesses mechanically interlocked with the microstructures on the pre-fabricated soft layer, securely anchoring it within the casting cavity. Subsequently, liquid silicone with a Shore hardness of 60A was cast into the mold. The assembly was then subjected to mechanical pressurization and elevated temperatures for thermal curing. This process ultimately yielded the soft inner layer part, a composite structure comprising two distinct silicone elastomers with disparate hardness levels. Supplementary Fig.S6c illustrates the fabrication of the soft encapsulation. The uncured 60A silicone is poured into heat-resistant resin part, after heating and curing, the fabricated encapsulation is removed.

Supplementary Fig. S7a depicts the assembly process of the designed F/T sensor, which can be divided into three steps. First, the PZT shell was aligned with the PCB and electrically connected using conductive silver paste (cured at 70 °C, 16 h) to avoid thermal depolarization, with epoxy resin infused into the gaps via capillary action to reinforce the ceramic. Second, a dip-coating method was used to apply adhesive to the microstructures of the soft inner layer (Supplementary Fig.S7b), which was then aligned and interlocked with the PZT electrodes using positioning features. Finally, the outer encapsulation was integrated using a 0.2 mm dimensional offset strategy to limit contact to the rigid support regions. This design protected the internal components while eliminating parasitic constraints, ensuring the free deformation of microstructures and high sensor sensitivity (More details can be found in Supplementary Note 2).

Fabrication process for annular PZT shells.

The subtractive process involves removing surface PZT material from initial workpieces through the interaction between the soft milling cutter and the workpieces, thereby achieving shell formation, where the soft cutter is fabricated with 95A TPU in a layered structure, enabling stiffness regulation through layer jamming. Typically, the fabrication of PZT shells using annular PZT workpieces (PZT, C-601, Fuji Piezo Ceramics, Inc.) mainly involves three stages: rough machining, finish machining, and polishing (Fig. 2c and Supplementary Fig. S2a). The soft machining system primarily consists of a robotic arm and a soft cutter. The robotic arm's displacement control system and end-effector force sensor are used to regulate milling parameters and monitor variations in milling forces, respectively. During the actual machining process, a multi-segment machining strategy is employed to mitigate the risk of workpiece breakage caused by excessive milling forces and stress concentrations. The approach ensures both a high success rate and machining efficiency. The machining process is segmented into five distinct phases with milling force feedback and thickness estimation (Supplementary Fig. S3). The first three segments correspond to the rough machining process, the fourth segment to the finishing process, and the final segment to the polishing process. And Supplementary Fig. S2b shows the process diagram. Once the target thickness is defined, the joint force sensors integrated into the robotic arm provide continuous real-time feedback on the milling force, which serves as a critical input to the machining system. If the estimated thickness of the PZT workpiece remains within the range of the current machining segment, the system continues to operate under the existing parameters. However, when the estimated thickness falls below the threshold of the current segment, the system automatically transitions to the next machining stage. Taking the machining process of a 20-mm-diameter annular PZT shell with a target thickness of 55 μm as an example (Fig. 2c), it can be divided into three stages at a milling speed of 110 rpm. During rough machining (0 to 1.2 h), a higher milling force is applied for rapid material removal from the initial 1-mm-thick PZT workpiece. As its thickness drops below 150 μm, the system transitions to finish machining (1.2 h to 1.5 h), reducing milling force to minimize stress concentration and preserve structural integrity. Finally, when the workpiece’s thickness reaches 60 μm, polishing begins (1.5 h to 2 h), replacing the milling layer with a polyurethane layer embedded with 3000-5000 mesh diamond abrasives.

Characterization of the machined annular PZT shells.

Images of the machined shells (Fig. 2d) and its surface morphology (Fig. 2e) were captured using a high depth of field camera and SEM. The overall morphology, surface details, and surface roughness of the electrodes were characterized using a high depth of field camera, SEM, and AFM (Supplementary Fig. S5b and S5c), respectively. For the dimensional precision of the machined annular PZT shells with various dimensions, the following measurements were conducted: the thickness was measured using a laser rangefinder LK-G80 from KEYENCE Co., Ltd.; the roundness was measured with a Mitutoyo Round-Test RA-1200 roundness meter from Mitutoyo Co., Ltd.; and the cylindricity was measured using a Zeiss ACCURA Coordinate Measuring Machine from Carl Zeiss AG Co., Ltd. The acoustic pressure characterization of the PZT-shell-based microphone involves sound signal generation, acoustoelectric signal conversion, and data acquisition. An arbitrary function generator (AFG3022C, Tektronix) generates sinusoidal signals, which are amplified by a power amplifier (5016, SAST) before driving a speaker (SUINY). A commercial microphone (MPA231, BSWA), positioned equidistant from the sound source as the PZT-shell microphone, serves as a reference for measuring incident sound pressure. A data acquisition system (DHDAS, Donghua) records signals from both the PZT-shell microphone and the reference microphone simultaneously. Sensitivity was determined by fitting sensor responses at different sound pressure levels using the least squares method. The entire test is conducted in a semi-anechoic chamber with a background noise level of 15.6 dB (Fig. 2f).

Design of the piezoelectric sensor metrology chain.

To guarantee high resolution and accuracy in the piezoelectric F/T sensing system, two distinct metrology chains were designed for different application scenarios. For calibration, characterization, and high-precision applications (Supplementary Fig. S9a), the tethered chain comprised an independent multi-channel charge amplifier (Type: KDT128AMP-072-AMP; KOST Technology Co., Ltd., Tianjin, China) as the analog front-end. The front-end features an electrometer-grade low input bias current of 75 fA and a nominal charge sensitivity of 10^{11} V/C. The conditioned signals were subsequently digitized by a 24-bit, 32-Ch synchronous data acquisition module (Type: DH5902N Rugged data acquisition system, Jiangsu Donghua Testing

Technology Co., Ltd.) serving as the digital back-end. This configuration provided an input-referred RMS noise floor of less than 40 μV at 1 ksps, which is significantly lower than the effective voltage LSB of less than 120 μV (detailed parameters are provided in Supplementary Table 5). Furthermore, to expand the practical application boundaries, we introduced a portable wireless metrology chain (Supplementary Fig. S9b, with detailed specifications in Supplementary Table 6) specifically designed for untethered scenarios, optimizing the balance between form factor, power, and precision. In this configuration, six sensing electrodes were sequentially routed to a single charge amplifier via a low leakage 8-channel CMOS multiplexer (MAX338, typical leakage 1 pA). To mitigate semiconductor-intrinsic charge injection (minimized to 1.5 pC by the MAX338) during switching, a dedicated settling time was implemented before each conversion. The conditioned output was then captured by a mixed-signal MCU (STM32G431), where the native 12-bit ADC resolution was enhanced to 14-bit via hardware oversampling. Crucially, the MCU leveraged its integrated DSP to perform on-chip edge computing—including Kalman filtering and real-time matrix decoupling—subsequently transmitting the finalized 6-axis load data via BLE for real-time monitoring. It should be noted that due to the intrinsic AC-coupled nature of the piezoelectric sensor metrology chain (which inherently acts as a high-pass filter with a low-frequency cutoff of 0.16 Hz), the sensor outputs reported in this study (denoted as ΔF, ΔT) effectively represent the dynamic force variations, rather than the absolute quasi-static total loads. This behavior is further demonstrated through a side-by-side comparison with a DC-coupled reference in Supplementary Fig. S9e

FEM analysis and validation.

We conducted multi-physics simulations in COMSOL to analyze the sensor's electromechanical behavior (Supplementary Fig. S8). The model consisted of an annular PZT shell with patterned outer electrodes at positive potential and the inner wall grounded, combined with elastomeric layers and a rigidly fixed base. Fasteners and minor fillets were omitted to reduce complexity without affecting local strain fields. The piezoceramic was modeled as linear anisotropic PZT using the strain-charge constitutive form, while elastomers were treated as nearly incompressible linear solids with moduli estimated from hardness correlations. The piezoelectric devices interface coupled solid mechanics and electrostatics, enabling both direct and converse effects. Mechanical loads were applied as distributed tractions (forces) or via rigid connectors (torques), reproducing F_x, F_y, F_z and M_x, M_y, M_z. Frequency-domain sweeps were carried out, with refined steps near resonances; convergence was confirmed when frequency shifts were <0.2% and electrode charge differences <1%. Quadratic tetrahedral meshes with 6–10 elements across the thinnest shell thickness ensured accuracy; mesh refinement was continued until terminal charge outputs varied by less than 2%. Simulation outputs were electrode charges or open-circuit voltages, assembled into a sensitivity matrix mapping six-component loads to six electrode signals. Overall, this workflow provided stable, consistent predictions of sensitivity and cross-axis coupling, closely aligned with experimental frequency responses.

Design of the sensor calibration and testing setup.

During the calibration and testing process, the designed F/T sensor was mounted on a precision 5-DOF displacement stage, featuring X-Y-Z translation, Z-axis rotation, and X-axis tilt (Models LT90-LM-2 and GCG60-35, Shenzhen Huike Pneumatic Precision Machinery Co., Ltd., China). To facilitate accurate calibration and benchmarking, a commercial six-axis force/torque sensor (Nano 17E, ATI Industrial Automation, USA) was mechanically connected in series with the proposed sensor. Dynamic mechanical loads were generated and applied by a voice coil motor (VCM) with a maximum peak force of 20 N (Model XZVC35-05A, Ningbo Smooth Motor Co., Ltd., China). The schematic of the experimental setup and the load adjustment mechanism is illustrated in Supplementary Fig. S10. The precision stage enables highly accurate alignment and spatial offset adjustments between the sensor axis and the centerline of the force application module, thereby fulfilling the requirements for applying various single-axis and combined complex loads. Further details regarding the testing apparatus can be found in the Supplementary Note 4.

Decoupling and calibration Process.

To derive an accurate dynamic decoupling matrix, calibration data was acquired using randomized, complex dynamic excitations (1–40 Hz) alongside a high-precision reference sensor (Supplementary Fig. S11). Crucially, a principal-axis-dominant loading strategy was employed to guarantee linear independence among the massive acquired voltage responses. For a specific frequency f, let L_{f} denote the true load matrix and V_{f} the voltage response matrix. To suppress high-frequency white noise, the local decoupling matrix C_{f} is derived by minimizing the Frobenius norm:

$$\min_{C_{\mathrm{f}}} \| L_{\mathrm{f}} - C_{\mathrm{f}} V_{\mathrm{f}} \|_{\mathrm{F}}^2 \tag{2}$$

This yields the analytical solution utilizing the Moore-Penrose pseudoinverse: $C_{\mathrm{f}} = L_{\mathrm{f}} V_{\mathrm{f}}^{+}$. To ensure broadband dynamic generalizability, block matrix concatenation is performed across all tested frequencies to form comprehensive matrices L_{global} and V_{global}. The global decoupling matrix C_{global} is then optimized by upgrading the objective function:

$$C_{\mathrm{global}} = \arg\min_{C} \left\| L_{\mathrm{global}} - C V_{\mathrm{global}} \right\|_{\mathrm{F}}^2 = L_{\mathrm{global}} V_{\mathrm{global}}^{+} \tag{3}$$

Furthermore, to mitigate residual non-linear errors ΔL, a secondary higher-order polynomial regression is introduced. Built upon the distinguishable strain patterns of our annular PZT configuration, the refined final sensor output model is formulated as

$$L_{\mathrm{pred}} = C_{\mathrm{global}} V + F_{\mathrm{comp}}(\mathrm{V}) \tag{4}$$

Where the explicit compensation function $F_{\mathrm{comp}}(\mathrm{V})$ expands into quadratic and cross-product terms governed by constant weight matrices W_{sq} and W_{cross}:

$$F_{\mathrm{comp}}(\mathrm{V}) = W_{\mathrm{sq}} V_{\mathrm{sq}} + W_{\mathrm{cross}} V_{\mathrm{cross}} \tag{5}$$

This mathematically transparent methodology efficiently mitigates cross-talk without relying on black-box neural networks.

Performance characterization of the designed F/T sensor.

To comprehensively evaluate the dynamic response capabilities of the designed F/T sensor, we systematically characterized the following core performance parameters. First, the dynamic sensitivity and cross-axis crosstalk of the sensor were tested (From Supplementary Figs. S12 to S32). Across the 1–40 Hz bandwidth, these were quantified by applying dominant-component randomized dynamic composite loads, followed by polynomial fitting via the least-squares method to extract sensitivity coefficients and quantify theoretical inter-axis interference. Second, the phase delay and dynamic hysteresis characteristics were characterized. Furthermore, the decoupling accuracy and absolute magnitude error under complex working conditions were strictly quantified. Based on 3,000 multi-dimensional composite load samples (Supplementary Table 7) per frequency point, incorporating randomized amplitudes, waveforms, and pre-loads, the decoupled outputs of the sensor were systematically compared against the ground truth of the commercial reference to calculate the load identification accuracy and the full-scale error percentage (%FS). Finally, the baseline noise under no-load conditions and minimum detectable signal of the sensor were measured without additional electromagnetic shielding. Detailed testing procedures and experimental conditions for each specific performance parameter are provided in Supplementary Note 4.

Effect of temperature variations on the sensor.

To evaluate the influence of ambient temperature (25 °C to 55 °C) on the sensor's thermo-mechanical coupling characteristics, a multi-level thermal characterization was conducted. At the material level, dynamic mechanical analysis (DMA) across a broadband (1–35 Hz) was utilized to extract the thermal evolution of the silicone's storage modulus and loss factor (Supplementary Fig. S33). At the device level, the thermal drift of dynamic sensitivity (Supplementary Fig. S34) and baseline noise variations (Supplementary Fig. S35) across different temperature gradients were systematically quantified via 1 Hz sinusoidal cyclic loading tests and multi-channel no-load voltage monitoring, respectively. Finally, for high-precision applications at elevated temperatures, we discussed potential compensation measures: utilizing a discrete-temperature calibration strategy to extract temperature-specific sensitivity compensation matrices at specific elevated temperature nodes (Supplementary Fig. S36). These matrices can directly replace the room-temperature decoupling matrix at the algorithmic level, thereby effectively suppressing thermally induced errors (More details can be found in Supplementary Note 5).

Characterization of batch manufacturing repeatability and performance consistency.

To verify the reliability and batch repeatability of the proposed manufacturing process, eight identically specified F/T sensors were fabricated and systematically characterized for their core performance consistency (More details can be found in Supplementary Note 6.). To ensure a standardized evaluation, all mechanical consistency tests were conducted under dynamic cyclic loading at 1 Hz. First, the absolute sensitivities across all load axes, and the cross-axis crosstalk matrices of each device in the batch were individually calibrated and statistically analyzed to quantify the baseline performance variance under identical manufacturing conditions (Supplementary Fig. S38). Furthermore, to evaluate the inter-device response consistency under varying thermal environments, the noise floor drift trajectories and dynamic sensitivity degradation curves of the eight sensors were extracted and compared across a temperature range of 25 °C to 55 °C (Supplementary Fig. S39). Through this multi-dimensional statistical characterization, the performance dispersion of the batch devices under dynamic decoupling, and complex thermo-mechanical coupling conditions was comprehensively quantified.

Application scenarios of the designed F/T sensor.

To demonstrate the exceptional integration portability and mechanical compatibility of the monolithic F/T sensor, we deployed it in multi-dimensional tactile sensing (Fig. 5) and intelligent exoskeleton monitoring (Fig. 6). Conventional metallic sensors are typically too bulky or rigid for such space- and weight-sensitive locations. In contrast, the proposed sensor features an ultra-lightweight profile (3.2 g) and soft silicone encapsulation, enabling seamless, conformal integration onto robotic fingertips for direct physical interaction without damaging target objects. For tactile applications, rather than tracking absolute quasi-static holding forces, we constructed an event-driven, closed-loop robotic arm-gripper control method. The control loop operates by detecting high-fidelity dynamic force/torque variations (ΔF, ΔT) captured during initial contact, collision impact, or incipient slip to indicate relative motion tendencies and execute discrete operational adjustments (e.g., precise assembly). More details can be found in Supplementary Note 7 and Supplementary Fig. S40.

For wearable exoskeleton applications, the sensor utilizes a customized internal "Soft Lid" architecture that enables direct mechanical coupling with rotating spindles and joint shafts, facilitating seamless embedment. To evaluate the intelligent exoskeleton for rehabilitation assessment, we deployed the device on wearers with diverse body morphologies and varying levels of lower-limb joint and muscle health. First, we monitored dynamic contact force variations (ΔF) between the limb and the exoskeleton during comfortable wear to infer appropriate donning tightness and provide personalized fit guidance. Additionally, the system quantifies relative dynamic variations across different motion processes by extracting transient peak joint torque variations (ΔT_{peak}) generated during distinct kinematic phases. These relative peak variations serve as objective metrics to track recovery progress (see Supplementary Note 8 and Supplementary Figs. S44 to S48 for testing and calibration details). For home-based function monitoring, a commercially available EMG sensor (Core Future Co., Ltd.) was affixed to the thigh to record quadriceps activity. Each test lasted 90 s, during which EMG signals and joint torque signals were recorded simultaneously. The transient peak torque variations and the EMG-to-torque time delay were subsequently obtained through data post-processing. The authors affirm that research participants provided informed consent for publication of the images in Fig.6d and Supplementary Video 4.

Data availability

All data supporting the findings of this study are available within the article and its supplementary files. Any additional requests for information can be directed to and will be fulfilled by the corresponding authors. Source data are provided with this paper.

Code availability

Data analysis made use of inbuilt functions in MATLAB and OriginLab. All parameters used for analysis are available in Methods and Supplementary Information.

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Author contributions

S.G. initiated the concept with the help of L.S. S.G. designed and performed all the experiments under the guidance of Z.Y., L.S., Y.Y., and W.Z., with help from X.W., Z.Z., and Q.Z., X.W. performed the FEM analysis. S.G., G.Z., S.Z., Z.D., and T.Z. designed the circuits. S.G. wrote the initial manuscript, which was revised by L.S., Y.Y., and Z.Y., Y.Y., Z.Y., L.S., and W.Z. supervised the project and provided funding. All authors contributed to the discussion and writing of this manuscript.

Competing interests

The authors declare no competing interests.

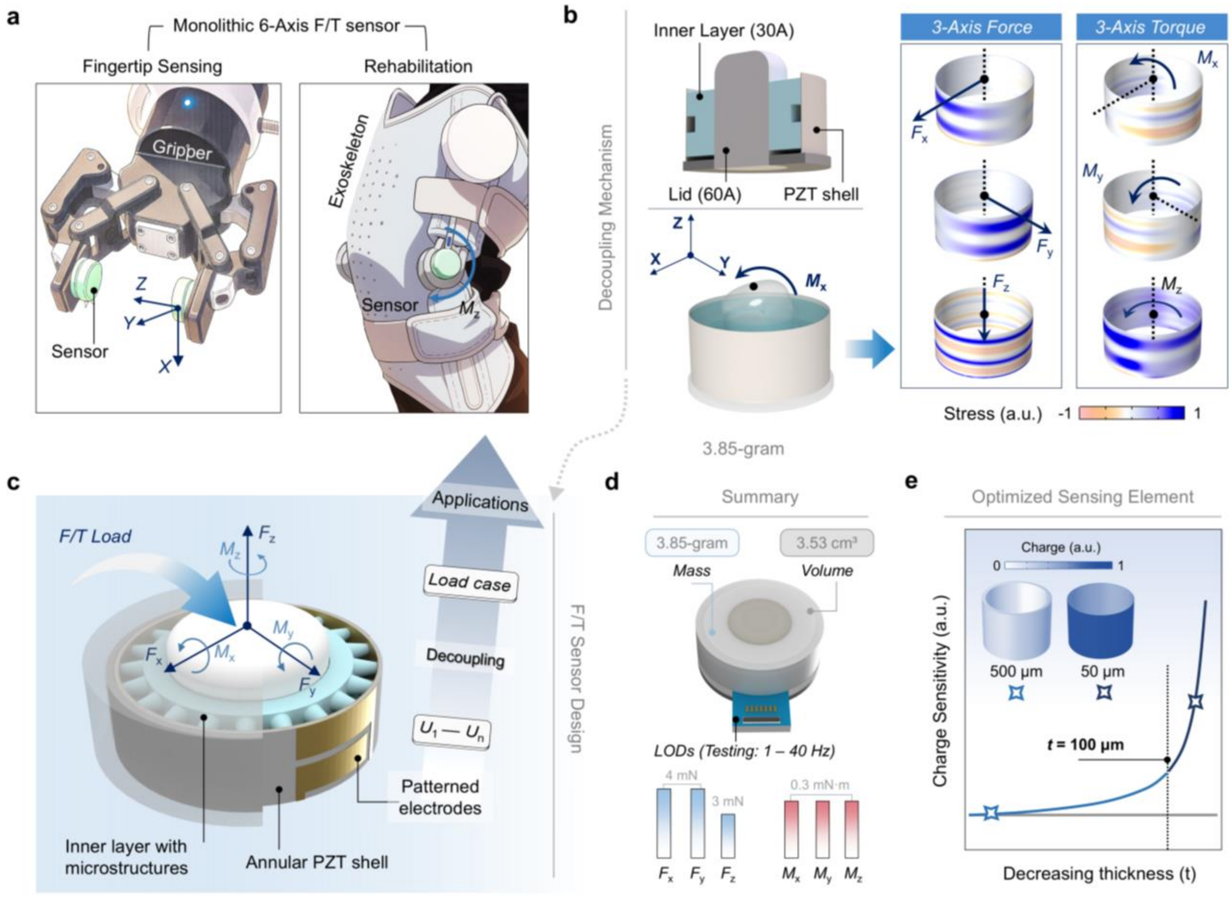

Fig. 1. General concept of the monolithic 6-axis F/T sensor. (a) Demonstration of F/T sensors integrated into robotic grippers and exoskeletons, enabling intelligence. (b) Schematic of the designed F/T sensing structure and stress distributions on the sensing element under different loads; the color scale represents the stress magnitude and type. (c) Design and decoupling process of the 6 axis F/T sensor, using an annular PZT shell as the sole sensing element and patterned electrodes to distinguish loads. (d) Key parameters of the designed F/T sensor. (e) Enhanced charge sensitivity of annular PZT shells with a thickness under 100 μm; the color scale represents the normalized charge amplitude.

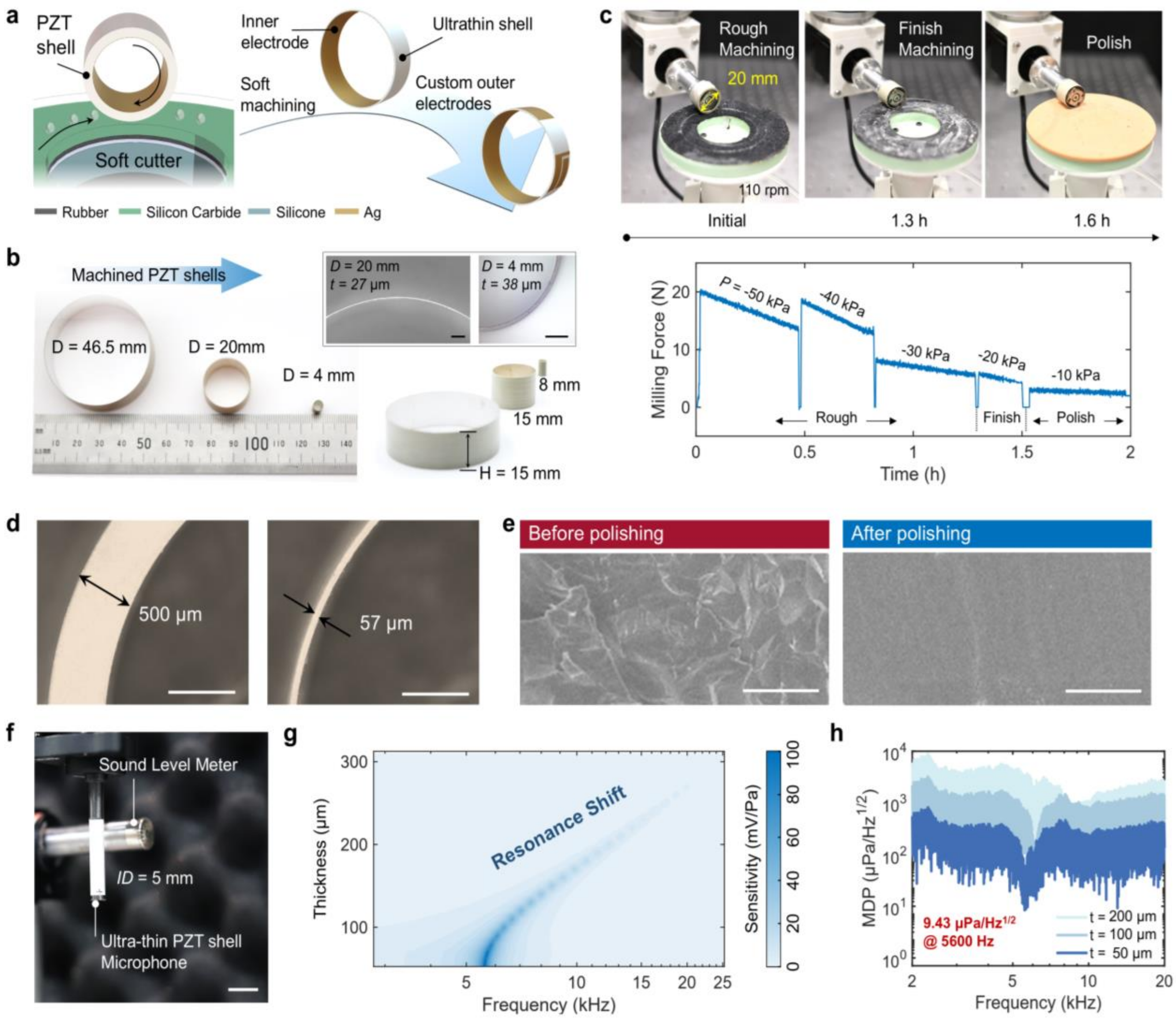

Fig. 2. Processing and characterization of annular PZT shells. (a) Illustration of the PZT shell-forming process. (b) Images of processed annular PZT thin shells of different dimensions, with uniform thickness. Scale bars 100 μm. (c) Photographs and monitored feedback milling force during processing of a 20-mm-diameter annular PZT shell with targeted 55-μm-thick. (d) Comparison of a processed 57-μm-thick annular PZT shell and a raw 500-μm-thick annular PZT shell. Scale bars 500 μm. (e) SEM images of shell surfaces before and after polishing. Scale bars 1 μm. (f) Schematic of the experimental setup, including a sound level meter for calibration. Scale bars 5 mm. (g) Measured voltage sensitivity of the PZT shells as a function of incident acoustic frequency and shell thickness; the color scale represents the amplitude of sensitivity. (h) Minimum detectable pressure (MDP) of annular PZT shells with different thicknesses as a function of

excitation frequency.

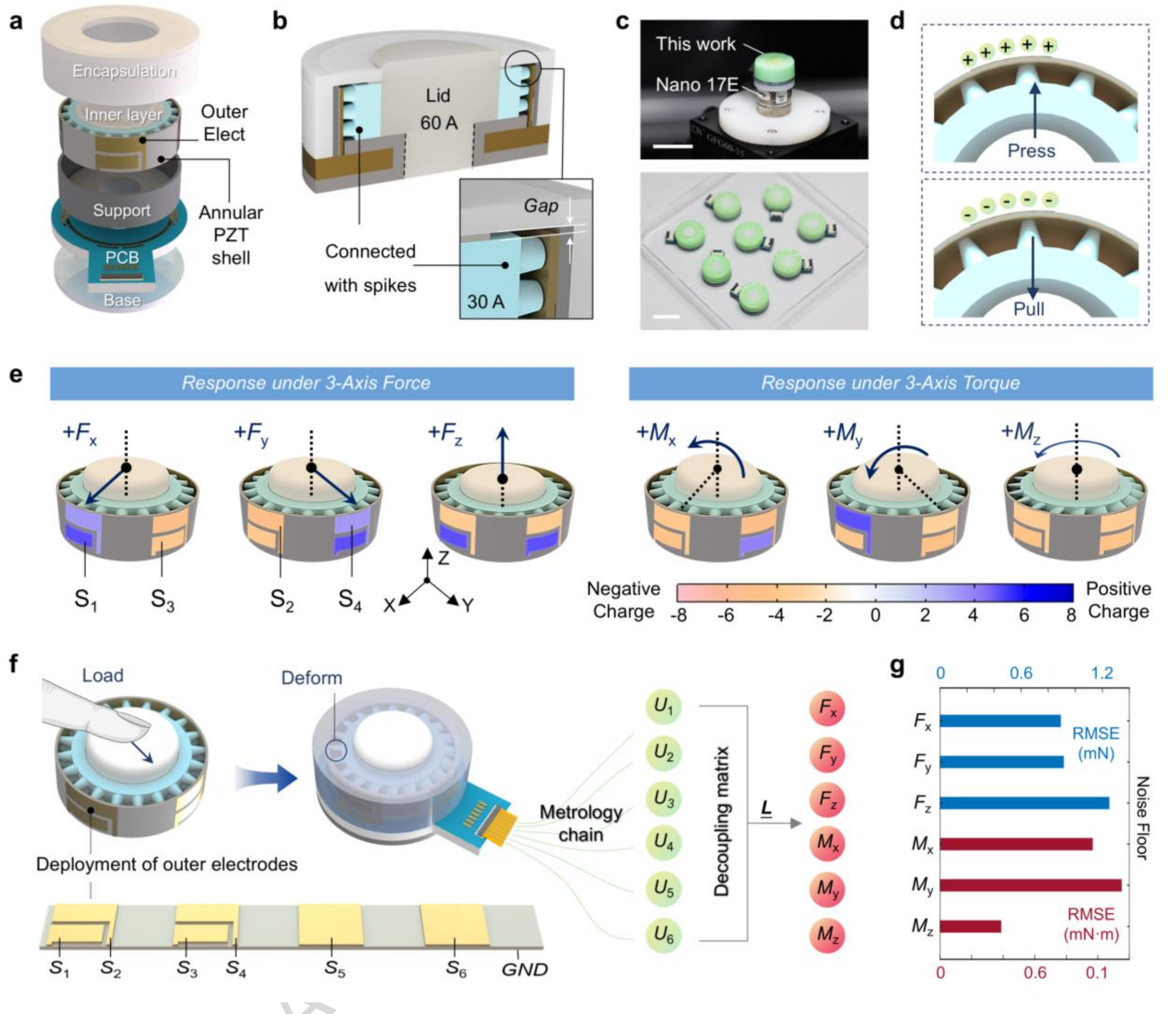

Fig. 3. Design and load decoupling of the 6-axis F/T sensor. (a) Exploded view of the designed F/T sensor. (b) Cross-section view of the the designed sensor, where a gap design is included between the encapsulation and soft inner layer. (c) Image of the the 6-axis F/T sensor. Scale bars 2 cm. (d) Charge outputs of the annular PZT shell subjected to pulling and pressing of the soft inner layer. (e) Schematic diagram of charge outputs of PZT shell external electrodes under pure single-axis F/T loads, taking electrodes S_1 to S_4 as examples; the color scale represents the charge magnitude and polarity. (f) Charge signal processing and load decoupling workflow of the F/T sensor, based on the six-electrode configuration on the annular PZT shell. (g) The baseline noise floor of each axis under no-load conditions.

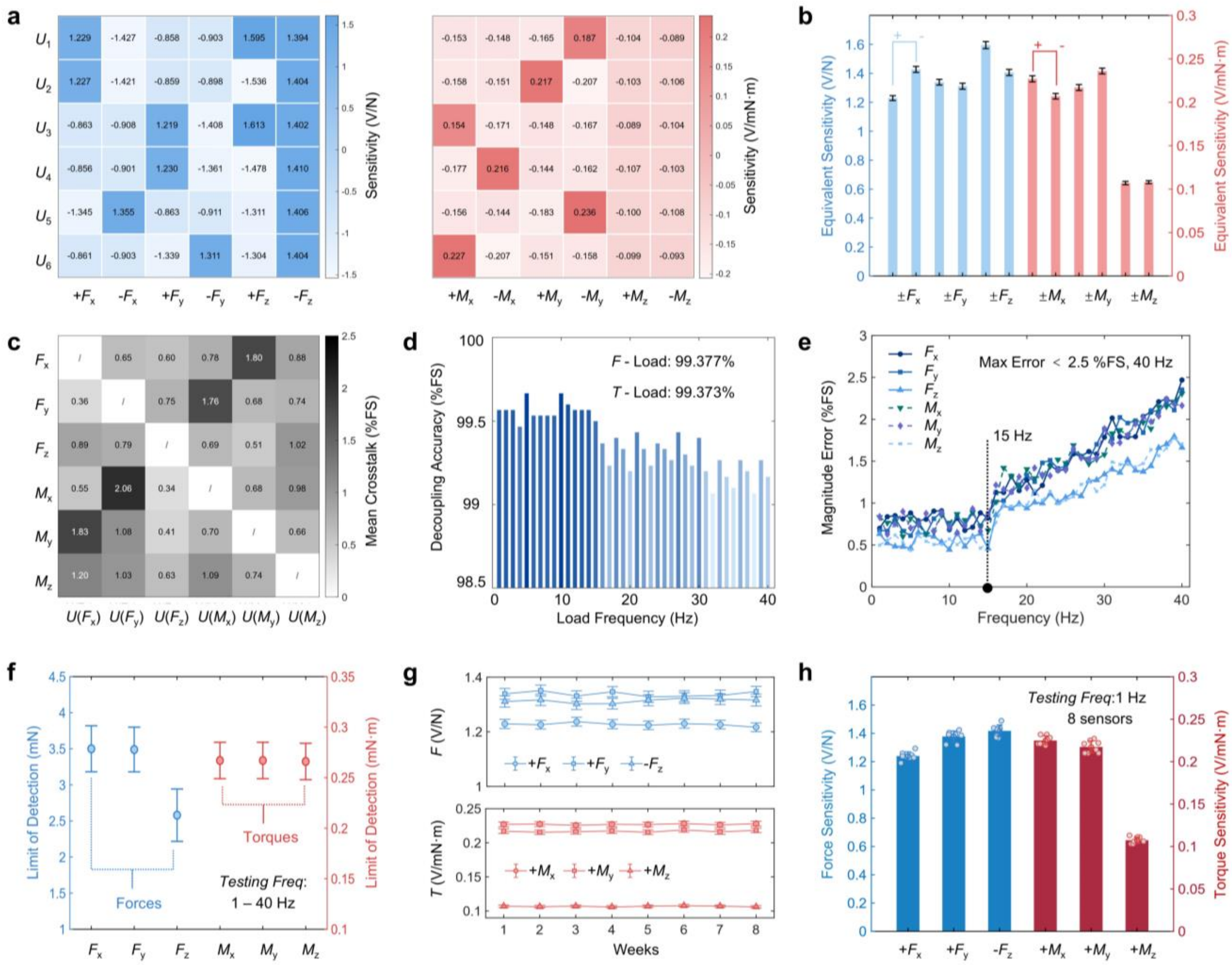

Fig. 4. Characterization of the 6-axis F/T sensor. (a) Measured average response sensitivity of the sensor electrodes to different loads under 1–40 Hz dynamic testing; the color scale represents the amplitude of sensitivity. (b) Variation range of the equivalent sensitivity for each load axis under 1–40 Hz dynamic loading. (c) Measured average cross-axis crosstalk of the sensor under 1–40 Hz dynamic tests; the color scale represents the amplitude of crosstalk, and bars show mean ± SD. (d) Measured decoupling accuracy of the sensor under 1–40 Hz dynamic loading. (e) Variation of the average amplitude error for each axis during 1–40 Hz dynamic tests. (f) Limit of detection of the sensor for 1–40 Hz dynamic loads; data are presented as mean ± SD. (g) Long-term stability test of the sensor, showing the variation of equivalent sensitivity for each axis over time; data are presented as mean ± SD. (h) Comparison of the sensitivity for each axis across eight consecutively fabricated sensors under 1 Hz dynamic loading; bars show mean ± SD.

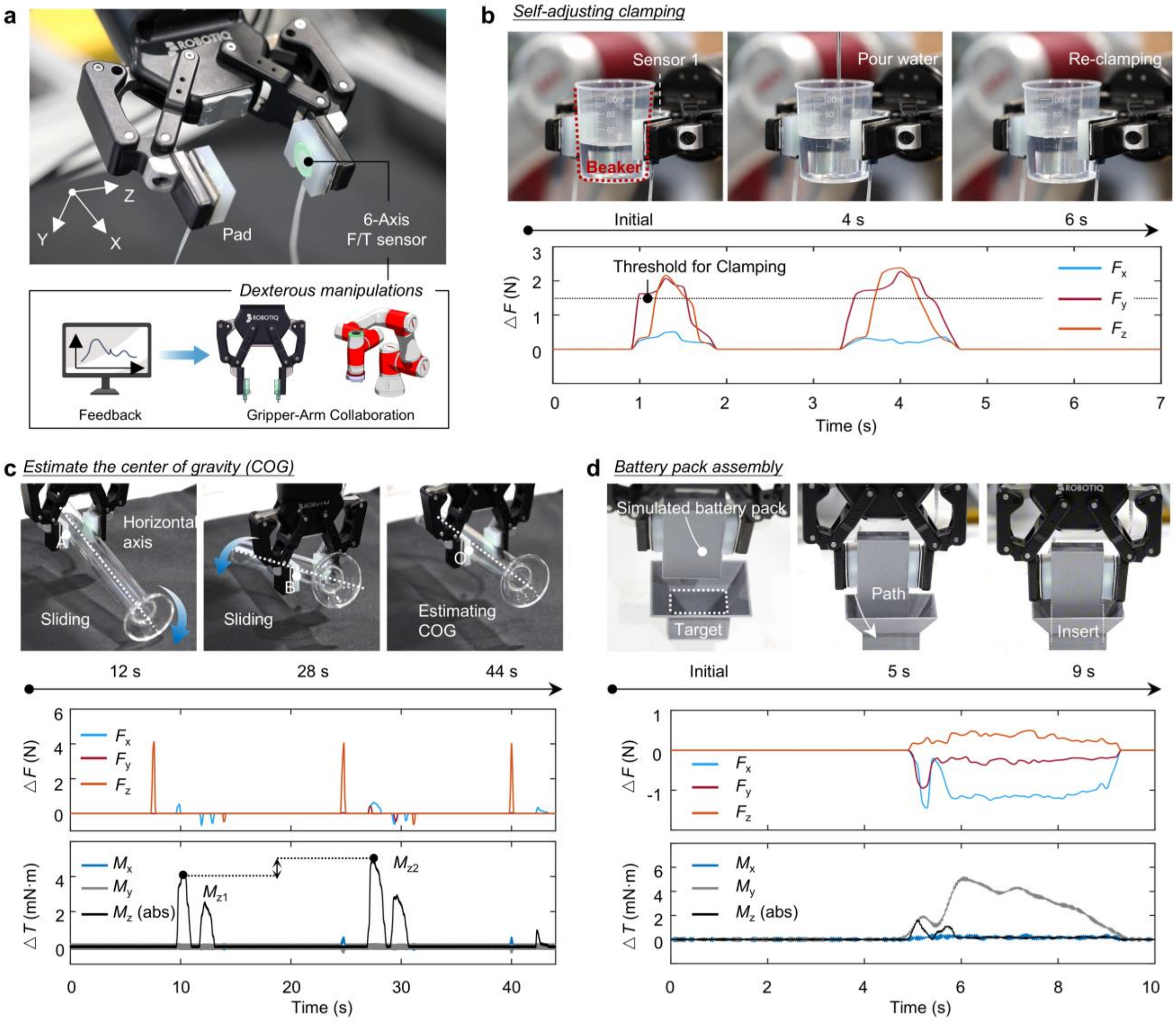

Fig. 5. F/T sensors enabling dexterous robotic gripper-arm collaboration. (a) Photograph of a robotic gripper with integrated sensors. (b) Experimental snapshots and monitored 3-axis force variations during a self-adjusting clamping task. (c) Experimental snapshots and monitored force/torque variations during a center of gravity (COG) estimation task on a transparent glass tube. (d) Experimental snapshots and monitored force/torque variations during a simulated battery pack assembly task. The computer element in Fig. 5a is created in BioRender. qu, J. (2026) https://BioRender.com/m87t8vp

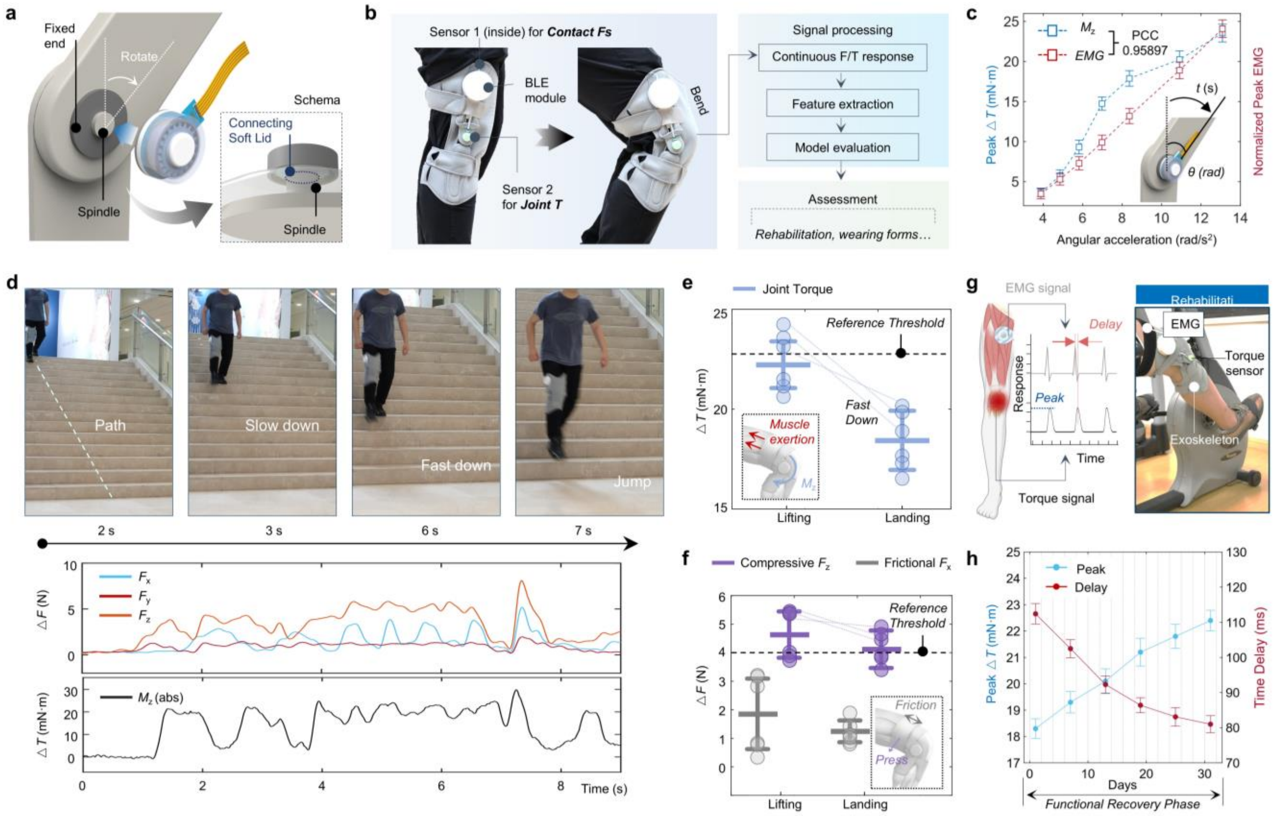

Fig. 6. F/T sensors enabling intelligent exoskeletons for motion monitoring and functional assessment. (a) Schematic diagram of the F/T sensor connected to the exoskeleton spindle via the inner layer. (b) Integration schematic of the intelligent exoskeleton equipped with two F/T sensors and its workflow for functional assessment. (c) Measured peak torque variations of sensor 2 and peak EMG signals of the quadriceps muscle as a function of knee joint angular acceleration; data are presented as mean ± SD. (d) Experimental snapshots and real-time monitored force/torque variations of a wearer during a stair-descent task. (e) Measured peak joint torque variations during the leg lifting and landing phases of the stair descent; bars show mean ± SD. (f) Measured peak compressive and frictional force variations between the wearer's leg and the exoskeleton during the corresponding phases; bars show mean ± SD. (g) Schematic illustration of the postoperative in-home rehabilitation assessment. (h) Measured peak knee joint torque variations and EMG-to-torque time delay as a function of exercise duration, quantitatively tracking the recovery; data are presented as mean ± SD. The graphical element in Fig. 6g is created in BioRender. qu, J. (2026) https://BioRender.com/m87t8vp